Near-strict 1-spike-per-cycle ceiling: 98.9% of (cell, cycle) pairs

exp046 · 4 June 2026

Abstract

exp041′s slope $p \approx 0.20$ was interpreted as *"each E cell joins a cycle with ≈ 20% probability"*. That reading assumes the per-cycle spike count is bounded by 1 (an E cell either participates in this cycle or not). This notebook measures that directly: walking through every gamma cycle in the shared 1,000-image held-out evaluation subset and counting how many spikes each E cell actually emits, on all 18 trained checkpoints in exp041′s τ_{GABA} sweep.

Methods

Cycle statistics are evaluated from the final-epoch checkpoints used by exp041, so this experiment audits the same endpoint gamma dynamics. The source training horizon is read from those checkpoint configurations and retained in the generated provenance.

For each of exp041′s 18 trained cells (6 τ_{GABA} × 3 seeds):

1. Run inference on the fixed 1,000-image subset of the official MNIST test partition; capture per-trial (T, B, N_E) and (T, B, N_I) spike tensors.
2. Detect I-burst times per trial: smooth the population I rate with a 1-ms Gaussian, run scipy peak detection with min-distance set to half the cell's own $1/f_\gamma$.
3. Cycle boundaries are the midpoints between consecutive I-burst peaks (first cycle starts at $t = 0$, last ends at trial end).
4. For each (cell, cycle, trial), count the number of E spikes within the cycle window.
5. Bucket counts globally into $\{0, 1, 2, \geq 3\}$ and aggregate by τ_{GABA}.

The cycle anchor is the I-burst: this is the right anchor because the cycle is operationally defined as *"the time between one inhibitory blanket and the next"*, not as the time between E bursts (which can be silent on a given cycle).

Results

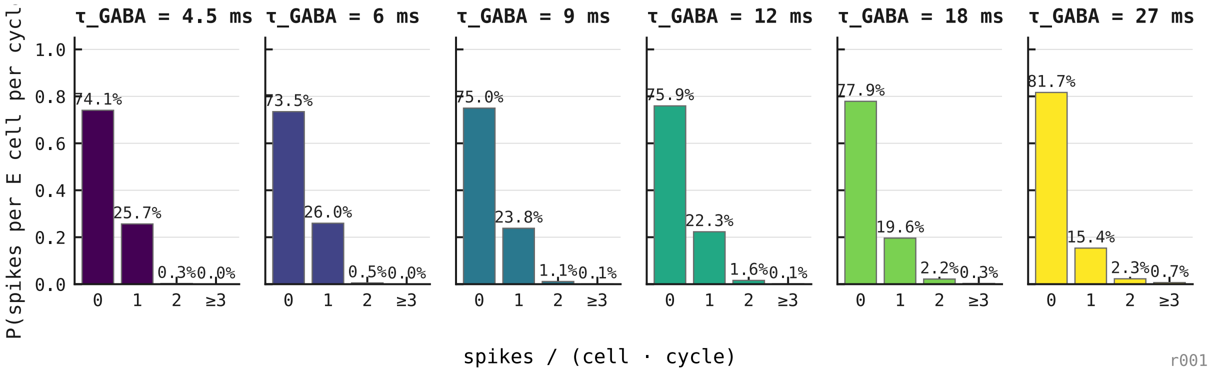

Figure 1: Distribution of E spike count per gamma cycle per cell, by τ_{GABA}, three seeds aggregated. Across **179 million (cell, cycle) pairs**, the architecture is overwhelmingly bimodal: each cell either emits zero spikes in a given cycle (≈ 79% of the time) or exactly one (≈ 20% of the time). Two-or-more events occur in ≈ 1.1% of cycles; three-or-more in ≈ 0.14%. Pooled over the sweep, 98.9% of pairs carry at most one spike.

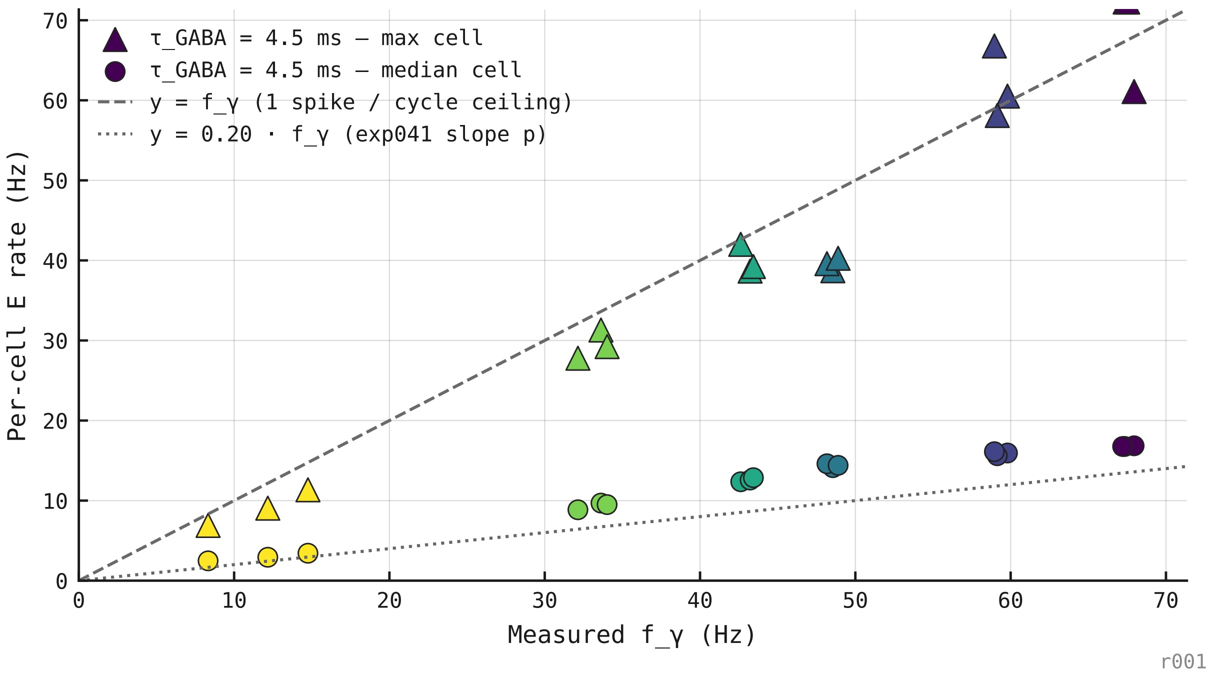

Figure 2: Per-cell E rate versus measured gamma frequency f_γ across the τ_{GABA} sweep. The busiest cell in each network tracks the one-spike-per-cycle ceiling $r = f_\gamma$ (max-cell fit $r = 0.97 f_\gamma$, $R^2 = 0.88$), while the median cell sits on exp041's shallower $r \approx 0.20 f_\gamma$ participation slope. The ceiling is near-strict: even the most active cell rarely exceeds one spike per cycle.